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Uranga et al. JAMA Intern Med 2016 five-day trial) and general standard medical
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share.google/b4bZ2uS3AdMDyhtPT link resolved to the WHO fact sheet. Chunks tagged
source GL are added general medical knowledge.",
    "disclaimer": "Educational information only. Not a substitute for professional
medical advice, diagnosis or treatment. Antibiotic choice and doses depend on local
resistance patterns, allergies, kidney and liver function, pregnancy, age and current
local guidelines. Doses listed are typical adult guideline doses and must be verified
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pus) in the small air sacs of the lungs (alveoli). It can be caused by bacteria, viruses or fungi. The immune system attacks the infection in the alveoli, which makes the lung tissue swell and leak fluid, making breathing difficult and limiting oxygen intake. It can cause fever and a cough with yellow, green or bloody mucus. Common causes include influenza (the flu), COVID-19 and pneumococcal disease. Treatment depends on the cause and the severity. Pneumonia can affect one lung or both lungs; pneumonia in both lungs is called bilateral or double pneumonia.",
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Pneumocystis jirovecii, Coccidioides) and protozoal infections such as toxoplasmosis.
Antibiotic-resistant bacteria such as MRSA (methicillin-resistant Staphylococcus
aureus), Klebsiella pneumoniae, Serratia marcescens and Pseudomonas aeruginosa can
cause hospital-related pneumonia. Gram-negative bacteria and Staphylococcus aureus
are typical in HAP and VAP. Pneumonia can also follow a viral infection (for example
Staphylococcus aureus after influenza).",
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pneumonia in children. Haemophilus influenzae type b (Hib) is the second most common
bacterial cause. Respiratory syncytial virus (RSV) is the most common viral cause. In
HIV-infected infants, Pneumocystis jirovecii is one of the most common causes,
responsible for at least a quarter of pneumonia deaths in HIV-infected infants.
Pneumonia can spread when viruses and bacteria normally found in the nose or throat
are inhaled into the lungs, via airborne droplets from coughs or sneezes, and through
the blood, especially during and shortly after birth.",
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that cause it are. Viral infections like flu, colds and COVID-19 can inflame the
lungs and are contagious, but most people with these infections do not develop
pneumonia. Some contagious infections, like pneumococcal disease, often cause
pneumonia. Fungal infections that cause pneumonia are not contagious from person to
person. For bacterial pneumonia, a person is generally no longer considered
contagious once the fever is gone and they have taken antibiotics for at least two
days. For viral pneumonia, a person is likely contagious until they feel better and
have been fever-free for several days.",
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      "keywords": ["chest X-ray", "CT scan", "sputum culture", "pulse oximetry", "ABG", "bronchoscopy", "pleural fluid", "blood tests", "urinary antigen", "PCR", "procalcitonin"],
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pneumococcal urinary antigen testing is recommended in severe CAP or in epidemiologic circumstances (such as a Legionella outbreak, recent travel). Influenza testing is recommended when influenza is circulating. Procalcitonin should not be used to decide whether to start antibiotics in adults with clinically suspected and radiographically confirmed CAP.",
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need brief observation, and classes IV to V usually require hospitalization. The 2019 ATS/IDSA criteria define severe CAP (ICU-level) as 1 major criterion or 3 or more minor criteria. Major: septic shock needing vasopressors; respiratory failure needing mechanical ventilation. Minor: respiratory rate 30 or more; PaO2/FiO2 ratio 250 or less; multilobar infiltrates; confusion/disorientation; uremia (BUN 20 mg/dL or more); leukopenia (WBC below 4000); thrombocytopenia (platelets below 100,000); hypothermia (below 36 C); hypotension requiring aggressive fluid resuscitation.",
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      "keywords": ["outpatient comorbidities", "amoxicillin clavulanate", "cefpodoxime", "levofloxacin", "moxifloxacin", "respiratory fluoroquinolone", "combination therapy"],
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      "content": "For hospitalized adults with non-severe CAP and no risk factors for MRSA or Pseudomonas: a beta-lactam PLUS a macrolide, or a beta-lactam PLUS doxycycline, or monotherapy with a respiratory fluoroquinolone. Beta-lactam options: ampicillin-sulbactam 1.5 to 3 g IV every 6 hours; cefotaxime 1 to 2 g IV every 8 hours; ceftriaxone 1 to 2 g IV daily; ceftaroline 600 mg IV every 12 hours. Macrolide options: azithromycin 500 mg daily or clarithromycin 500 mg twice daily. Fluoroquinolone options: levofloxacin 750 mg daily, moxifloxacin 400 mg daily.
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Patients can be switched to oral therapy once clinically stable and able to take oral medication.",
      "keywords": ["inpatient CAP", "ceftriaxone", "ampicillin-sulbactam", "beta-lactam plus macrolide", "hospitalized pneumonia treatment", "IV to oral switch"],
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      "content": "For severe CAP (ICU-level) without MRSA or Pseudomonas risk factors, recommended empirical therapy is a beta-lactam (ampicillin-sulbactam, cefotaxime, ceftriaxone or ceftaroline) PLUS a macrolide (strong recommendation, low-quality evidence), or a beta-lactam PLUS a respiratory fluoroquinolone (strong recommendation, low-quality evidence). Severe patients need monitoring, oxygen support (high-flow nasal oxygen, non-invasive or invasive ventilation as needed), IV fluids, and management of sepsis if present. Corticosteroids: the 2019 ATS/IDSA guideline recommended against routine use in non-severe CAP and suggested against routine use in severe CAP; however, more recent trials (for example CAPE COD, 2023, hydrocortisone 200 mg/day) and subsequent critical-care guidance support considering corticosteroids in severe CAP, so this decision should follow current local protocols and clinician judgement.",
      "keywords": ["severe CAP treatment", "ICU pneumonia", "corticosteroids pneumonia", "hydrocortisone", "high-flow oxygen", "mechanical ventilation", "sepsis"],
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      "content": "Extend empirical coverage only when risk factors are present. For MRSA risk (prior respiratory isolation of MRSA, or recent hospitalization with IV antibiotics in the last 90 days plus locally validated risk factors): add vancomycin (about 15 mg/kg IV every 12 hours, adjusted by levels/AUC) or linezolid 600 mg every 12 hours, and obtain cultures and MRSA nasal PCR to allow de-escalation. For Pseudomonas risk: add an anti-pseudomonal beta-lactam such as piperacillin-tazobactam 4.5 g IV every 6 hours, cefepime 2 g every 8 hours, ceftazidime 2 g every 8 hours, imipenem 500 mg every 6 hours, meropenem 1 g every 8 hours, or aztreonam 2 g every 8 hours (with appropriate additional agents). Stop MRSA or Pseudomonas coverage if cultures and PCR are negative and the patient improves.",
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      "keywords": ["duration of antibiotics", "5 days", "7 days", "clinical stability", "how long antibiotics pneumonia", "short course", "Uranga"],
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    {
      "id": "tx-009",
      "section": "Treatment",
      "title": "Antiviral treatment (influenza, COVID-19 and others)",
      "content": "Providers usually do not treat viral pneumonia with specific medication, but may prescribe antivirals to shorten illness and reduce severity. Influenza: oseltamivir (typically 75 mg twice daily for 5 days in adults) is recommended for patients with suspected or confirmed influenza pneumonia and for hospitalized patients, ideally started as early as possible (benefit is greatest within 48 hours but treatment is still recommended in severe or hospitalized cases). COVID-19: options include nirmatrelvir/ritonavir (5 days, for eligible non-hospitalized patients at high risk, with attention to drug interactions) and remdesivir (IV); hospitalized hypoxemic patients may also receive corticosteroids such as dexamethasone per current guidelines. RSV and other viral causes are managed supportively in most adults. Antibiotics are added only if bacterial co-infection is suspected.",
      "keywords": ["antivirals", "oseltamivir", "Tamiflu", "COVID-19 pneumonia treatment", "nirmatrelvir", "remdesivir", "viral pneumonia treatment", "influenza pneumonia"],
      "audience": "clinician",
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      "id": "tx-010",
      "section": "Treatment",
      "title": "Antifungal treatment for fungal pneumonia",
      "content": "Antifungal medications treat pneumonia caused by fungi. Treatment depends on the fungus: Pneumocystis jirovecii pneumonia (PJP) is treated with trimethoprim-sulfamethoxazole (typically 15 to 20 mg/kg/day of the trimethoprim component in divided doses for 21 days), with adjunctive corticosteroids when hypoxemia is significant; Aspergillus infection with voriconazole or isavuconazole;
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endemic fungi such as Coccidioides (Valley fever) with fluconazole or itraconazole in mild cases and amphotericin B in severe disease; Cryptococcus with amphotericin B plus flucytosine followed by fluconazole. Fungal pneumonia is not contagious person to person. Specialist (infectious disease) input is recommended.",
      "keywords": ["antifungal", "fungal pneumonia treatment", "Pneumocystis", "PJP", "TMP-SMX", "voriconazole", "fluconazole", "amphotericin B", "Valley fever"],
      "audience": "clinician",
      "source": "CC, GL"
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      "id": "tx-011",
      "section": "Treatment",
      "title": "Treatment by specific pathogen",
      "content": "Streptococcus pneumoniae: amoxicillin or penicillin-based therapy (ceftriaxone if hospitalized), choice adjusted to susceptibility. Mycoplasma pneumoniae and Chlamydia pneumoniae (atypical): macrolide, doxycycline, or respiratory fluoroquinolone; beta-lactams are ineffective because these organisms lack a cell wall. Legionella: levofloxacin or azithromycin, typically 7 to 14 days (longer in immunocompromised patients). Haemophilus influenzae: amoxicillin-clavulanate or a second/third-generation cephalosporin. Methicillin-susceptible Staphylococcus aureus: anti-staphylococcal beta-lactam such as cefazolin or nafcillin; MRSA: vancomycin or linezolid. Pseudomonas aeruginosa: anti-pseudomonal beta-lactam (sometimes with a second agent). Influenza: oseltamivir. Pneumocystis: trimethoprim-sulfamethoxazole.",
      "keywords": ["pathogen-specific treatment", "atypical pneumonia treatment", "Legionella treatment", "Mycoplasma treatment", "pneumococcal treatment", "Staphylococcus aureus pneumonia"],
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      "id": "tx-012",
      "section": "Treatment",
      "title": "Aspiration pneumonia treatment",
      "content": "Aspiration pneumonia occurs when food, liquid, saliva or vomit enters the lungs. Treatment includes antibiotics, supportive care and addressing the cause of aspiration (swallowing evaluation, positioning, feeding modification). Under current ATS/IDSA guidance, anaerobic coverage should not be routinely added for suspected aspiration pneumonia unless lung abscess or empyema is suspected. Typical regimens when anaerobic coverage is needed include ampicillin-sulbactam or amoxicillin-clavulanate; clindamycin or metronidazole combined with other agents may be used in specific situations. Lung abscess generally requires prolonged antibiotics (often several weeks) guided by imaging and clinical response.",
      "keywords": ["aspiration pneumonia treatment", "anaerobic coverage", "ampicillin-sulbactam", "lung abscess", "swallowing evaluation", "dysphagia"],
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      "section": "Treatment",
      "title": "Hospital-acquired and ventilator-associated pneumonia treatment",
      "content": "HAP and VAP are usually more serious because resistant bacteria such as MRSA and Pseudomonas aeruginosa are more likely. IDSA/ATS 2016 guidance recommends empirical therapy that covers Staphylococcus aureus and Pseudomonas and other gram-negative bacilli, tailored to local antibiograms and individual risk factors (for example prior IV antibiotics, septic shock, ARDS, high local MRSA prevalence), with antibiotics narrowed once culture results return. A 7-day course is generally recommended. Prevention measures include elevating the head of the bed, daily sedation review, oral care, subglottic secretion drainage, minimizing ventilator days, and hand hygiene.",
      "keywords": ["HAP treatment", "VAP treatment", "hospital pneumonia antibiotics", "7 days", "antibiogram", "ventilator pneumonia prevention"],
      "audience": "clinician",
      "source": "CC, GL"
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      "id": "tx-014",
      "section": "Treatment",
      "title": "Oxygen therapy, fluids and respiratory support",
      "content": "Oxygen therapy: if blood oxygen is low, extra oxygen is given through a nasal tube or face mask. Targets are usually oxygen saturation of about 92 to 96% in most adults (88 to 92% in people with COPD and carbon dioxide retention risk). WHO recommends oxygen for children when saturation is below 90%. Severe respiratory failure may need high-flow nasal oxygen, non-invasive ventilation (CPAP/BiPAP) or mechanical ventilation. IV fluids treat or prevent dehydration, given directly into a vein. Chest physiotherapy, deep-breathing exercises, incentive spirometry and early mobilization help clear secretions and recover lung function.",
      "keywords": ["oxygen therapy", "oxygen saturation target", "IV fluids", "high-flow oxygen", "CPAP", "ventilation", "breathing exercises"],
      "audience": "general",
      "source": "CC, WHO, GL"
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      "section": "Treatment",
      "title": "Draining fluid around the lungs (pleural effusion and empyema)",
      "content": "If a lot of fluid collects around the lungs (parapneumonic pleural effusion), a provider may drain it with a catheter or surgery. Pleural fluid is sampled to check for infection; if it is pus (empyema), acidic or infected, drainage via a chest tube is needed, sometimes with fibrinolytic medicines or surgical clearing (video-assisted thoracoscopic surgery, VATS) if the fluid is loculated. Antibiotics continue throughout and may be needed for several weeks in empyema.",
      "keywords": ["pleural effusion", "empyema", "chest tube", "drain fluid", "VATS", "parapneumonic effusion"],
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      "section": "Treatment",
      "title": "When hospitalization is needed",
      "content": "Severe pneumonia may require hospital treatment. Hospitalization is more likely in people under age 2, over age 65, with a weakened immune system, or with heart and lung conditions. Clinical indicators for admission include low oxygen saturation, rapid breathing, low blood pressure, confusion, inability to take oral fluids or medication, high CURB-65 or PSI scores, multilobar involvement, or complications such as effusion or sepsis. Most hospitalized patients start IV antibiotics and switch to oral once stable.",
      "keywords": ["hospital admission", "when to hospitalize", "severe pneumonia", "who needs hospital", "inpatient criteria"],
      "audience": "general",
      "source": "CC, GL"
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      "id": "tx-017",
      "section": "Treatment",
      "title": "At-home treatment and symptom relief",
      "content": "If the provider says it is safe to recover at home: take prescribed antibiotics exactly as directed and finish the course; use pain relievers and fever reducers such as ibuprofen (Advil) or acetaminophen (Tylenol) for aches and fever; use prescribed breathing treatments and exercises to loosen mucus; run a humidifier or take steamy showers or baths; drink plenty of fluids to loosen mucus and stay hydrated; rest. Check with a healthcare provider before taking cough suppressants, because coughing helps clear the lungs. Avoid smoking and smoke exposure. Children should not be given aspirin.",
      "keywords": ["home treatment", "home remedies", "ibuprofen", "acetaminophen", "humidifier", "fluids", "cough suppressants", "rest", "recover at home"],
      "audience": "general",
      "source": "CC, GL"
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    {
      "id": "tx-018",
      "section": "Treatment",
      "title": "Antibiotic safety, side effects and stewardship",
      "content": "Common antibiotic side effects: nausea, diarrhea, rash and yeast infections. Antibiotic-associated diarrhea can occasionally be Clostridioides difficile infection, which needs medical review if diarrhea is severe or persistent. Seek urgent care for signs of severe allergy (swelling of face or throat, hives with breathing difficulty). Macrolides and fluoroquinolones can prolong the QT interval; fluoroquinolones carry boxed warnings for tendon, nerve and CNS effects. Doxycycline should be avoided in pregnancy and in young children when alternatives exist. Antibiotics should be used only for bacterial infection and for the shortest effective duration to limit resistance.",
      "keywords": ["antibiotic side effects", "C. difficile", "allergy", "QT prolongation", "fluoroquinolone warnings", "antibiotic resistance", "stewardship"],
      "audience": "general",
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      "id": "tx-019",
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      "title": "No improvement: treatment failure",
      "content": "If symptoms are not improving after 48 to 72 hours of treatment, or worsen, the provider will reassess: confirm adherence and absorption, review the diagnosis, consider resistant or unusual organisms (MRSA, Pseudomonas, Legionella, fungi, tuberculosis), complications (pleural effusion, empyema, lung abscess), non-infectious mimics (heart failure, pulmonary embolism, lung cancer), and may repeat imaging, obtain cultures, widen or change antibiotics, or admit to hospital. Prompt medical review is important for anyone who is getting worse.",
      "keywords": ["not improving", "treatment failure", "worsening pneumonia", "no response to antibiotics", "reassess"],
      "audience": "clinician",
      "source": "GL"
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      "id": "tx-020",
      "section": "Treatment",
      "title": "Pneumonia in pregnancy: treatment considerations",
      "content": "Pregnancy increases the risk and severity of pneumonia, especially influenza pneumonia and COVID-19. Treatment typically uses a beta-lactam (amoxicillin, ampicillin, ceftriaxone) with azithromycin when atypical coverage is needed. Doxycycline and fluoroquinolones are generally avoided. Oseltamivir is recommended for suspected influenza in pregnancy. Oxygen saturation should be kept at or above about 95% to ensure adequate fetal oxygenation. Pregnant patients with pneumonia should be assessed promptly by a clinician.",
      "keywords": ["pregnancy pneumonia", "pregnant treatment", "azithromycin pregnancy", "avoid doxycycline pregnancy", "oseltamivir pregnancy"],
      "audience": "clinician",
      "source": "CC, GL"
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      "id": "tx-ped-001",
      "section": "Treatment - children",
      "title": "Antibiotic treatment of pneumonia in children (WHO)",
      "content": "WHO states that pneumonia in children should be treated with antibiotics. The first-line antibiotic is amoxicillin dispersible tablets. Most cases require oral antibiotics, often prescribed at a health centre, and can also be diagnosed and treated at community level by trained community health workers with inexpensive oral antibiotics. Hospitalization is recommended only for severe cases. Dosing is by child weight and age according to national or WHO protocols and must be confirmed with a health professional. In many national guidelines, fast-breathing or chest-indrawing pneumonia is treated with oral amoxicillin for about 3 to 5 days. Severe pneumonia (danger signs) needs referral and injectable antibiotics, oxygen and supportive care.",
      "keywords": ["child pneumonia treatment", "amoxicillin dispersible tablets", "WHO treatment", "community health workers", "oral antibiotics children", "severe pneumonia children"],
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      "source": "WHO, GL"
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      "id": "tx-ped-002",
      "section": "Treatment - children",
      "title": "WHO danger signs and referral in children",
      "content": "Signs of very severe or severe disease in a child with cough or difficult breathing include inability to drink or breastfeed, convulsions, lethargy or unconsciousness, vomiting everything, stridor in a calm child, severe malnutrition, central cyanosis, oxygen saturation below 90%, severe respiratory distress with head nodding, and hypothermia in young infants. These children need urgent referral to hospital for injectable antibiotics (for example ampicillin or benzylpenicillin plus gentamicin), oxygen and supportive care. The WHO and UNICEF Global Action Plan for Pneumonia and Diarrhoea (GAPPD) emphasizes that every sick child should be able to get the right antibiotics and oxygen to get well.",
      "keywords": ["danger signs", "child referral", "severe pneumonia in children", "oxygen children", "injectable antibiotics", "GAPPD"],
      "audience": "child",
      "source": "WHO, GL"
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    {
      "id": "tx-ped-003",
      "section": "Treatment - children",
      "title": "HIV-related pneumonia prophylaxis in children",
      "content": "In children infected with HIV (and HIV-exposed infants), WHO recommends daily cotrimoxazole to decrease the risk of pneumonia, including Pneumocystis jirovecii pneumonia. Pneumocystis jirovecii is responsible for at least one quarter of pneumonia deaths in HIV-infected infants.",
      "keywords": ["cotrimoxazole", "HIV children", "prophylaxis", "Pneumocystis infants"],
      "audience": "child",
      "source": "WHO"
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    {
      "id": "tx-ped-004",
      "section": "Treatment - children",
      "title": "Pediatric CAP in high-income settings",
      "content": "In settings following IDSA/PIDS or AAP guidance, previously healthy, fully immunized children with suspected bacterial CAP are treated with high-dose amoxicillin (commonly around 90 mg/kg/day divided in two doses, with the exact dose and duration set by the pediatrician; shorter courses of about 5 days are increasingly supported by trials). Atypical pneumonia in school-aged children is treated with a macrolide such as azithromycin. Influenza pneumonia is treated with oseltamivir. Hospitalized children needing IV therapy typically receive ampicillin or ceftriaxone. Bacterial pneumonia in children is not routinely treated with cough or cold medicines.",
      "keywords": ["pediatric CAP", "high-dose amoxicillin", "azithromycin children", "AAP", "IDSA PIDS", "child antibiotics duration"],
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      "source": "GL"
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      "title": "Complications of pneumonia",
      "content": "Serious complications include: fluid around the lungs (pleural
effusion), which may become infected (empyema); bacteria in the bloodstream
(bacteremia) leading to sepsis and organ failure; lung abscess; acute respiratory
distress syndrome (ARDS) and respiratory failure. Other possible complications
include septic shock, cardiac events (arrhythmia, heart attack, worsening heart
failure), kidney injury, and meningitis in pneumococcal disease. Older adults and
people with chronic conditions face higher risks of complications.",
      "keywords": ["complications", "sepsis", "ARDS", "respiratory failure", "pleural
effusion", "lung abscess", "bacteremia", "empyema"],
      "audience": "general",
      "source": "CC, GL"
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      "id": "comp-002",
      "section": "Complications",
      "title": "Treating complications",
      "content": "Sepsis requires early recognition, IV antibiotics, IV fluids, and
sometimes vasopressors in the ICU. ARDS and respiratory failure need oxygen support,
lung-protective mechanical ventilation and intensive care. Pleural effusion and
empyema need drainage (chest tube, sometimes fibrinolytics or surgery). Lung abscess
usually needs prolonged antibiotics (often 4 to 6 weeks or longer) and sometimes
drainage. Any complication is a reason for urgent medical care.",
      "keywords": ["treat sepsis", "ARDS treatment", "empyema drainage", "lung
abscess treatment", "complication management"],
      "audience": "clinician",
      "source": "GL, CC"
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      "section": "When to seek care",
      "title": "When to see a healthcare provider",
      "content": "Contact a healthcare provider if you have new or worsening
shortness of breath, high fever, cough with mucus, tiredness, or unexplained loss of
appetite. Pneumonia can be life-threatening if left untreated, especially in people
with underlying health conditions, infants, and older adults. Children with fast
breathing, chest indrawing or poor feeding should be assessed promptly.",
      "keywords": ["when to see a doctor", "seek care", "warning signs", "medical
attention", "red flags"],
      "audience": "general",
      "source": "CC, WHO"
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      "id": "when-002",
      "section": "When to seek care",
      "title": "Emergency warning signs",
      "content": "Go to the emergency room or call emergency services (911 in the US)
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if you struggle to breathe or are short of breath while sitting still, have new or worsening chest pain, or are confused or cannot think clearly. Other emergency signs: bluish lips or fingertips, severe weakness, fainting, very rapid heartbeat, coughing up significant blood, inability to keep fluids down, or in a child unable to drink or breastfeed, convulsions, or unusual drowsiness.",
      "keywords": ["emergency", "call 911", "ER", "cyanosis", "severe shortness of breath", "confusion", "chest pain emergency"],
      "audience": "general",
      "source": "CC, WHO"
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      "section": "Prognosis and recovery",
      "title": "How soon will I feel better?",
      "content": "Recovery time depends on age, the cause, how sick the patient is, and whether there are other health conditions or complications. Most symptoms of bacterial pneumonia start to improve within a day or two of starting treatment. Cough and fatigue can linger for several weeks. If hospitalized, it may take six to eight weeks to feel back to normal. Fever typically resolves within 2 to 3 days of effective treatment in most uncomplicated cases.",
      "keywords": ["recovery time", "how long pneumonia lasts", "prognosis", "feel better", "lingering cough", "fatigue", "six to eight weeks"],
      "audience": "general",
      "source": "CC"
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      "id": "prog-002",
      "section": "Prognosis and recovery",
      "title": "Long-term effects and follow-up",
      "content": "After recovery some people notice trouble with physical activity (shortness of breath or reduced stamina), worsening of cardiovascular disease, and a general decline in quality of life. Children who have recovered from pneumonia have an increased risk of chronic lung diseases. Patients should follow up with their provider if concerns persist after recovery. The 2019 ATS/IDSA guideline does not recommend routine follow-up chest imaging in adults whose symptoms have resolved within 5 to 7 days, although some guidelines suggest a follow-up chest X-ray after about 6 to 8 weeks for smokers and people over 50 to exclude an underlying lung abnormality such as a tumor.",
      "keywords": ["long-term effects", "follow-up", "follow-up chest X-ray", "post-pneumonia", "stamina", "chronic lung disease"],
      "audience": "general",
      "source": "CC, GL"
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      "section": "Prognosis and recovery",
      "title": "Can pneumonia go away on its own?",
      "content": "Viral pneumonia often goes away on its own, but a healthcare provider should always evaluate suspected pneumonia, because some cases are serious, especially when bacteria cause it. Bacterial pneumonia is best treated with
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antibiotics. The only way to know the difference is to see a provider.",
      "keywords": ["go away on its own", "untreated pneumonia", "self-resolve",
"walking pneumonia"],
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      "source": "CC"
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      "id": "prev-001",
      "section": "Prevention",
      "title": "Vaccines that reduce pneumonia risk",
      "content": "The best way to prevent pneumonia is vaccination against the
bacteria and viruses that commonly cause it. These vaccines do not protect against
all types of pneumonia, but make severe illness less likely. Pneumococcal vaccines:
several vaccines exist, each recommended for certain age groups or risk groups (for
example conjugate vaccines PCV15, PCV20 and PCV21 for adults and PCV15/PCV20 in
childhood schedules; recommended ages for adults vary by country and have been
lowered to 50 in the US). Influenza vaccine annually. COVID-19 vaccines. RSV vaccines
for older adults and certain at-risk groups and RSV immunization for infants
(maternal vaccine or monoclonal antibody). Childhood vaccines including Hib,
pneumococcal conjugate, measles and pertussis (whooping cough) vaccines. Local
schedules vary, so a provider should advise which vaccines apply.",
      "keywords": ["vaccines", "pneumococcal vaccine", "PCV20", "flu shot", "COVID
vaccine", "RSV vaccine", "Hib", "measles", "pertussis", "prevention"],
      "audience": "general",
      "source": "CC, WHO, GL"
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      "id": "prev-002",
      "section": "Prevention",
      "title": "Everyday precautions",
      "content": "Quit smoking and avoid secondhand smoke (smoking damages the lungs
and increases infection risk). Do not vape (vaping increases the risk of lipoid
pneumonia). Wash hands with soap and water before eating, before and after handling
food and after using the restroom; use alcohol-based hand sanitizer if soap is
unavailable. Avoid close contact and sharing items with people who have contagious
illnesses such as the flu, colds or COVID-19. In hospitals or care facilities, ask
providers how to reduce infection risk. Treat other infections and health conditions
that weaken immunity. Avoid excessive alcohol. Good dental hygiene and safe
swallowing practices reduce aspiration risk.",
      "keywords": ["prevent pneumonia", "hand washing", "quit smoking", "vaping",
"avoid alcohol", "hygiene", "secondhand smoke"],
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      "source": "CC, GL"
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      "id": "prev-003",
      "section": "Prevention",
      "title": "Prevention in children (WHO)",
      "content": "Immunization against Hib, pneumococcus, measles and whooping cough
(pertussis) is the most effective way to prevent childhood pneumonia. Adequate
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nutrition improves natural defences, starting with exclusive breastfeeding for the first 6 months of life, which also helps shorten illness. Reducing indoor air pollution (for example with affordable clean indoor stoves) and encouraging good hygiene in crowded homes reduce pneumonia cases. HIV-infected children receive daily cotrimoxazole prophylaxis. The WHO/UNICEF GAPPD strategy has three arms: protect (exclusive breastfeeding and adequate complementary feeding), prevent (vaccination, handwashing with soap, reducing household air pollution, HIV prevention and cotrimoxazole prophylaxis) and treat (access to community or facility care with antibiotics and oxygen).",
      "keywords": ["child pneumonia prevention", "exclusive breastfeeding", "immunization", "clean stoves", "GAPPD", "protect prevent treat"],
      "audience": "child",
      "source": "WHO"
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      "id": "sp-001",
      "section": "Special populations",
      "title": "Older adults",
      "content": "Pneumonia is a leading cause of hospitalization and death in older adults. Presentation may be atypical (confusion, falls, weakness, poor appetite, no fever). Aspiration is more common with swallowing problems or dementia. Recovery is slower and complications such as cardiac events, delirium and functional decline are more common. Vaccination (pneumococcal, influenza, COVID-19, RSV), good oral care and prompt medical review of new symptoms are especially important.",
      "keywords": ["elderly", "older adults pneumonia", "delirium", "aspiration", "vaccination elderly"],
      "audience": "general",
      "source": "CC, GL"
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      "id": "sp-002",
      "section": "Special populations",
      "title": "Immunocompromised patients",
      "content": "People with weakened immune systems (HIV, cancer chemotherapy, organ or stem cell transplant, long-term steroids or biologic drugs) are at higher risk of severe pneumonia and of unusual organisms, including Pneumocystis jirovecii, fungi (Aspergillus, Cryptococcus), cytomegalovirus and mycobacteria. They need prompt evaluation, often CT imaging and bronchoscopy, and specialist input. Symptoms may be mild or subtle.",
      "keywords": ["immunocompromised", "HIV pneumonia", "transplant", "chemotherapy", "opportunistic infection", "PJP"],
      "audience": "clinician",
      "source": "CC, GL"
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      "id": "sp-003",
      "section": "Special populations",
      "title": "Infants and young children",
      "content": "Children under 2 are at higher risk of hospitalization. Infants may show fast breathing, grunting, nasal flaring, chest indrawing, poor feeding,
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irritability or lethargy rather than a typical cough. RSV is the most common viral cause; Streptococcus pneumoniae and Hib are the leading bacterial causes. Prevention relies on routine vaccination, breastfeeding, avoiding smoke exposure and good hygiene. Parents should seek care urgently if a child has fast or labored breathing or cannot feed.",
      "keywords": ["infants", "toddlers", "children pneumonia", "grunting", "nasal flaring", "RSV", "poor feeding"],
      "audience": "child",
      "source": "CC, WHO, GL"
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      "section": "FAQ",
      "title": "FAQ: Is pneumonia curable?",
      "content": "Yes. Most people recover fully with appropriate treatment. Bacterial pneumonia is treated with antibiotics and usually begins to improve within one to two days; viral pneumonia often resolves with supportive care; fungal pneumonia is treated with antifungal medicines. Recovery can take weeks, and severe cases or people with other illnesses may need hospital care.",
      "keywords": ["is pneumonia curable", "can pneumonia be cured", "full recovery"],
      "audience": "general",
      "source": "CC, WHO"
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      "section": "FAQ",
      "title": "FAQ: What is the best treatment for pneumonia?",
      "content": "There is no single best treatment; it depends on the cause and severity. For uncomplicated bacterial community-acquired pneumonia in healthy adults, guideline first-line options include amoxicillin or doxycycline (or a macrolide where resistance is low). For adults with other conditions or in hospital, combination therapy such as a beta-lactam plus macrolide or a respiratory fluoroquinolone is used. Viral pneumonia is mainly supportive care (fluids, rest, fever control, oxygen if needed) with antivirals for influenza or COVID-19 when indicated. Fungal pneumonia is treated with antifungals. In children, WHO recommends amoxicillin dispersible tablets as first line. Severe disease needs hospital care with oxygen and IV therapy.",
      "keywords": ["best treatment", "what medicine for pneumonia", "which antibiotic", "treatment options summary"],
      "audience": "general",
      "source": "CC, WHO, GL"
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      "id": "faq-003",
      "section": "FAQ",
      "title": "FAQ: Can pneumonia be treated at home?",
      "content": "Many mild cases of community-acquired pneumonia in otherwise healthy adults and children can be treated at home with oral antibiotics (if bacterial), rest, fluids and fever control, but only when a healthcare provider has
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assessed the patient and said it is safe. Hospital care is more likely for people under 2, over 65, with weakened immunity, with heart or lung disease, or with low oxygen levels, confusion, rapid breathing or inability to take oral medication.",
      "keywords": ["treat pneumonia at home", "outpatient", "home care", "mild pneumonia"],
      "audience": "general",
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      "section": "FAQ",
      "title": "FAQ: Should I take cough medicine for pneumonia?",
      "content": "Check with a healthcare provider before taking cough suppressants, because coughing helps clear mucus and infection from the lungs. Fluids, a humidifier or steam, and rest can help ease the cough. Over-the-counter pain and fever relievers such as acetaminophen or ibuprofen can be used as advised.",
      "keywords": ["cough suppressant", "cough medicine", "mucus", "should I suppress cough"],
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      "source": "CC"
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      "title": "FAQ: What is double (bilateral) pneumonia?",
      "content": "Double or bilateral pneumonia means the infection affects both lungs. It is not a separate disease and is not necessarily more dangerous in every case, but involvement of multiple lobes (multilobar pneumonia) is a marker of severity and can cause more breathing difficulty.",
      "keywords": ["double pneumonia", "bilateral pneumonia", "both lungs", "multilobar"],
      "audience": "general",
      "source": "CC, GL"
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      "section": "FAQ",
      "title": "FAQ: What is walking pneumonia?",
      "content": "Walking pneumonia is an informal term for a mild form of pneumonia, often caused by Mycoplasma pneumoniae, in which a person feels unwell but can still go about daily life. Typical symptoms include a persistent dry cough, low-grade fever, sore throat, headache and fatigue. It is commonly treated with a macrolide such as azithromycin or with doxycycline, and sometimes resolves with rest, though evaluation by a provider is advised.",
      "keywords": ["walking pneumonia", "Mycoplasma pneumoniae", "mild pneumonia", "atypical pneumonia"],
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      "id": "faq-007",
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      "title": "FAQ: How can I tell if pneumonia is getting better?",
      "content": "Signs of improvement include fever resolving, easier breathing, reduced chest pain, better appetite and energy, and decreasing cough or mucus over days to weeks. Cough and fatigue can persist for several weeks. Worsening breathlessness, returning fever after improvement, new chest pain, confusion or coughing up blood require prompt medical review.",
      "keywords": ["pneumonia recovery signs", "getting better", "improving", "worsening", "relapse"],
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      "section": "FAQ",
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      "content": "COVID-19 and influenza are viral infections that can lead to viral pneumonia, but most people with these infections do not develop pneumonia. Pneumonia describes inflammation and fluid in the lungs from any infectious cause; COVID-19 and flu are specific diseases that may or may not involve the lungs. Pneumonia may also be caused by bacteria or fungi. Testing (viral PCR or antigen tests plus chest imaging) distinguishes them.",
      "keywords": ["COVID vs pneumonia", "flu vs pneumonia", "viral pneumonia", "differences"],
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      "source": "CC, GL"
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    {
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      "section": "FAQ",
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      "content": "No. Antibiotics do not work against viruses. A provider may still prescribe them if a bacterial infection is present at the same time as a viral infection, or if bacterial co-infection cannot be ruled out in a sick patient. Viral pneumonia is generally managed with supportive care, and antivirals when appropriate (for example oseltamivir for influenza).",
      "keywords": ["antibiotics virus", "viral pneumonia antibiotics", "do antibiotics work", "co-infection"],
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    {
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      "section": "FAQ",
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      "content": "Pneumonia can be serious and even life-threatening, especially for babies and young children, adults over 65, people with chronic illnesses or weakened immune systems, and if treatment is delayed. WHO reports it as the single largest infectious cause of death in children worldwide. With timely diagnosis and
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appropriate treatment, most healthy people recover fully.",
      "keywords": ["is pneumonia serious", "dangerous", "life-threatening", "death",
"mortality"],
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    {
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      "section": "Glossary",
      "title": "Glossary of pneumonia terms",
      "content": "Alveoli: tiny air sacs in the lungs where oxygen enters blood.
Consolidation: lung tissue filled with fluid or pus, appearing as an area of opacity
on imaging. Empyema: collection of pus in the pleural space. Pleural effusion: fluid
between the lung and chest wall. Cyanosis: bluish discoloration of skin, lips or
nails from low oxygen. Hypoxemia: low blood oxygen. Sputum: mucus coughed up from the
lungs. Empirical therapy: treatment started on the basis of the most likely cause
before test results are available. Aspiration: inhaling food, liquid or stomach
contents into the airways. ARDS: severe lung injury causing respiratory failure.
Sepsis: life-threatening organ dysfunction caused by a dysregulated response to
infection. Antibiotic stewardship: using antibiotics only when needed, at the right
dose and duration. MRSA: methicillin-resistant Staphylococcus aureus.",
      "keywords": ["glossary", "definitions", "terms", "alveoli", "consolidation",
"empyema", "cyanosis", "hypoxemia"],
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